

ML Specification Report — playground-series-s3e14

Wild Blueberry Yield Prediction · our from-scratch agent (GBDT pool + tree-search v3)

Figures grounded in the competition's `facts.json`, `eda_summary.json`, `STATUS.md`, and `experiments_tree_v3.json`.

Overview

The task is to predict wild-blueberry `yield` from field/environmental measurements (metric: MAE, lower is better). We solve it with a from-scratch pipeline — a pool of gradient-boosted trees (LightGBM / XGBoost / CatBoost plus a regularized LGB) refined by linear iteration and a 60-node harness-v3 tree search that ends on a 34-member "kitchen-sink" mega-blend. Our submission-ready linear champion scores **CV OOF MAE 340.59891** (≈ 340.6), and the tree-search research champion (node #44, OOF-only) reaches a lower **340.35572**. The NVIDIA reproduce-agent, copying a public first-place kernel, reaches **337.28** — NVIDIA ahead by ~ 3.3 MAE; its true edge was OOF-stacking, which we did not reproduce.

Why it matters. Yield prediction from pollination and weather variables informs planting, pricing, and resource allocation in specialty-crop agriculture.

Data

Purpose of Data. Predict the continuous `yield` of wild-blueberry plots — a tabular regression Playground Series episode (Aygün et al. Nature 2026 benchmark, Season 3 Episode 14). **Data Format** is clean **tabular CSV**, entirely numeric. **Data Volume** is **15,289 training rows / 10,194 test rows**, with **16 raw predictive features** once `id` and the target are removed (there are **no categorical columns**).

Data Quality is high: **no missing values** in either split. The sources disagree on exact-duplicate rows — `eda_summary.json` reports **14 duplicate rows** in train while `STATUS.md` reports **7 exact duplicate rows** (both describe them as harmless); the discrepancy is noted rather than resolved. Train $\leftrightarrow$ test means differ by **<1% on every column** (largest shift `RainingDays` -0.82%), so there is no covariate shift. The strongest EDA signals, all consistent with blueberry-biology intuition, were:

- **Fruit-biology block dominates:** `fruitset` ($r = +0.886$), `seeds` ($r = +0.869$), `fruitmass` ($r = +0.826$) are by far the strongest predictors — prime candidates for interaction/product terms.
- **Rain suppresses yield:** `RainingDays` ($r = -0.477$) and `AverageRainingDays` ($r = -0.484$) are strongly negative; the two are near-collinear ($r = 0.991$).
- `clonesize` is a moderate negative discrete driver ($r = -0.383$, only 6–8 distinct values).
- **Temperature columns are redundant and weak:** the six Upper/Lower TRange columns are near-perfectly collinear (pairwise $r \geq 0.998$) yet individually near-zero-signal ($|r| \approx 0.02$) — condensed and later mostly dropped.

- **Pollinator columns** (`honeybee` , `bumbles` , `andrena` , `osmia`) are individually weak ($|r|$ 0.07–0.20); `honeybee` is extremely right-skewed (skew $\approx$ 41.6).

Annotation Guidelines. The label is `yield` $\in \mathbb{R}$, a continuous float over [1945.53, 8969.40] (mean 6025.19, std 1337.06, skew -0.29 , only 776 distinct values but **not** integer-valued — `is_integer_valued` = `false`, `log_transform_candidate` = `false`). Submissions are **real-valued yield predictions** scored by MAE.

Feature Set. From the 16 raw columns we engineer up to **27 features** (iteration 1), later pruned to a **21-feature** working set (iteration 2, after a LGB gain-importance probe dropped near-zero features), grouped below:

Group	Features
Raw (kept)	<code>clonesize</code> , <code>honeybee</code> , <code>bumbles</code> , <code>andrena</code> , <code>osmia</code> , <code>fruitset</code> , <code>fruitmass</code> , <code>seeds</code> , <code>RainingDays</code> , <code>AverageRainingDays</code> , <code>MaxOfUpperTRange</code> , <code>MinOfUpperTRange</code> , <code>MaxOfLowerTRange</code>
Fruit-biology interactions	<code>fruitset_x_seeds</code> , <code>fruitset_x_fruitmass</code> , <code>fruitmass_x_seeds</code> , <code>fruit_triple</code> , <code>seeds_per_fruitset</code>
Temperature (condensed)	<code>temp_spread</code> , <code>temp_avg</code> (<code>temp_avg</code> dropped in iter 2)
Pollinator index	<code>total_pollinators</code> , <code>honeybee_share</code>
Rain / scale	<code>rain_intensity</code> (dropped iter 2), <code>log_clonesize</code> (dropped iter 2)
Dropped noise (iter 2)	<code>temp_avg</code> , <code>log_clonesize</code> , <code>rain_intensity</code> , <code>MinOfLowerTRange</code> , <code>AverageOfLowerTRange</code> , <code>AverageOfUpperTRange</code>

Splitting strategy. A single canonical **5-fold KFold CV (shuffle, seed = 42)**, shared across every linear iteration and every tree-search node so scores are directly comparable. Stratification is unnecessary — the target is continuous and i.i.d. with no groups/time structure and no train $\leftrightarrow$ test shift (validation hint: *"continuous i.i.d. target $\rightarrow$ standard KFold"*). The Kaggle leaderboard was not used (no credentials in this environment), so **all scores here are CV-OOF only**; no public/private LB numbers exist for this run.

Models & Architecture

Purpose of Architecture. A regression model that minimizes MAE on wild-blueberry yield. **Architecture Type** is a **gradient-boosted decision-tree ensemble** — LightGBM as the primary learner, with XGBoost, CatBoost (native categorical handling of the low-cardinality env columns), and a regularized LGB member, combined by a weighted blend plus snap-to-grid post-processing. The submission-ready champion is a **5-member simplex blend** (linear exp #7); the overall lowest-OOF champion is the **34-member harness-v3 mega-blend** discovered at tree-search node #44.

Each member takes the same **Input Format** — a purely numeric feature matrix (no encoding needed; CatBoost optionally treats the low-cardinality env columns as native categoricals) — of **Input Dimension 21 features** per row in the working configuration (a 1-D vector; no spatial/sequence structure).

Architecture Description. Each GBDT member is an additive ensemble of decision trees grown by histogram-based gradient boosting under an MAE (`regression_l1`) objective. The blend layer sits on top as a convex weighted average of the members' out-of-fold predictions, with weights found by grid-simplex search (linear iterations) or by v3's k=800 Dirichlet sampling + coordinate-ascent search (tree search); snap-to-grid is applied to the blended output. **Model Complexity** is best expressed as trees $\times$ leaves rather than a dense parameter count — a representative tuned LGB member used `num_leaves` = 63, `max_depth` = 5 (Optuna box, exp #4); exact per-member tree/leaf counts for the root members are **not recorded**. The submission-ready champion blends 5 such members; the tree-search champion blends 34, of which 28 retained weight > 0.005.

Training procedures

Training proceeds as a staged progression on the shared 5-fold split, each stage measured on the same OOF scale:

Stage	Configuration	OOF MAE
Baseline (exp #1)	generic 3-model LGB+XGB+CAT blend (16 raw features)	341.40782
Iteration 1 (exp #2)	27 features + fruit-biology interactions, 3-model blend + snap-to-grid	340.95856
Iteration 2 (exp #3)	pruned to 21 features + native-cat CatBoost, grid-simplex blend + snap (beat Ridge stack 344.04)	340.71180
Phase B (exp #5)	+ Optuna-tuned LGB added as 4th member, 4-way simplex + snap	340.62702
Phase B (exp #7, linear champion)	+ seed-2024 LGB as 5th member, 5-way simplex + snap	340.59891
Tree search v3 (exp #8, node #44)	34-member explore-burst mega-blend, k=800 weight search + snap (OOF-only)	340.35572

The **Loss Function** is mean absolute error (LightGBM `regression_l1` / MAE objective), with model selection directly on OOF MAE. The **Optimization Algorithm** is gradient boosting (histogram-based) — **not** SGD/ADAM — with blend weights optimized by grid-simplex (linear) and Dirichlet + coordinate-ascent (tree v3). The **Learning Rate** is the boosting shrinkage; the Optuna-tuned LGB used `learning_rate` = 0.0154 (exp #4), while the root members' learning rates are **not recorded** individually. There is **no Learning Rate Scheduler** (N/A — *GBDT convergence is governed by CV early-stopping, not a schedule*) and **no Batch Size** (N/A — *full-dataset histogram boosting, not mini-batched*).

Training Duration for the tree-search v3 run was ~2,338 s wall (~39 min) over 60 node evaluations (the hard budget cap); ~12 min of that was wasted on 6 catastrophic DART evals (solo OOF ~6,100+ MAE) that contributed nothing. The linear Phase-B rounds were ~3–6 min each; the Optuna LGB tune was 36 fold-0-proxy trials in 293 s. **Training Memory** was **not recorded** — a 15k $\times$ 21 float matrix is trivial on the single arm64 CPU machine, peak not separately measured. **Transfer Learning** is N/A (*no pretrained weights*); its analogue here is **cross-season experience injection** — s3e7's "don't re-tune the second model, add tuned variants to the pool" lesson was carried in and directly shaped Phase B. **Data Augmentation** is N/A (*tabular*); the analogues are feature engineering (the 21–27 features) and seed-bagging.

Reproducibility Standards are strict: a fixed 5-fold KFold (shuffle, seed = 42) shared across all scripts and the tree, with solo-node OOF scores digit-verified against the STATUS ledger to 5 decimals (root asserted == 342.02154). The tree-search champion came from a **native harness-v3 run — no replay needed** (`v3_native = true`); the run resumes deterministically from `experiments_tree_v3.json` at seed 42. Blend weight search is deterministic at seed 42. Honest caveat carried in-source: the 34-dim mega-blend weights were fit on the full OOF with no nested validation, so the burst-beats-exploit *direction* is robust but the 5th decimal is not.

Inference procedures

Decision Threshold is *N/A* — this is regression, not classification, and the target is not integer-valued (`is_integer_valued = false`), so no thresholding or rounding is applied. The one output transform is **snap-to-grid**: final predictions are snapped to the **nearest observed train yield value**. This was validated empirically (the EDA noted the target is *not* a coarse integer grid, so snapping was tested rather than assumed) and won consistently by a small margin — e.g. raw blend 340.65207 → snapped 340.59891 in the linear champion, and raw 340.57853 → snapped 340.35572 in the tree-search champion (where snap-to-grid was applied inside the metric function on every candidate weight vector). **Inference Duration** was **not recorded**; GBDT scoring of the 10,194 test rows is sub-second per member on CPU. **Inference Memory** is negligible and **not recorded** (tree inference is lightweight). Note the tree-search champion is an **OOF-only research result** — no test predictions were generated for it, so it has no submission file; the submission-ready deliverable is the linear 5-way blend (340.59891).

Evaluation & Benchmarking

Task for Performance Evaluation. Predict continuous wild-blueberry yield and minimize MAE. **Performance Metrics.** MAE is the sole competition metric; our submission-ready linear champion reaches **CV OOF MAE 340.59891** ($\approx$ **340.6**, the reported `our_best`), improving 341.40782 → 340.95856 → 340.71180 → 340.62702 → 340.59891 across the linear stages, and the tree-search v3 mega-blend pushes further to **340.35572** (OOF-only). No leaderboard scores exist — this run was never submitted to Kaggle.

Performance Benchmarking. The natural comparator is the **NVIDIA reproduce-agent**, which copies the strongest public kernel:

Agent	Approach	CV MAE	Note
NVIDIA	reproduces <code>sergiosaharovskiy/ps-s3e14-first-place-winning-solution</code> (core)	337.28	winner
Our agent	from-scratch GBDT pool + tree-search v3	340.6	-3.3 MAE

NVIDIA leads by $\sim$ 3.3 MAE. The honest caveat: the public kernel's real edge was **OOF-stacking**, which our reproduction did **not** capture (`repro_type = core`) — so this is a comparison of a well-optimized public first-place solution against our from-scratch pool, and the gap here reflects that missing stacking layer rather than a difference in base-model quality. On this S3-era yield problem the well-optimized public kernel

retains a clear edge; the reproduce-vs-originate gap narrows on newer seasons, analyzed separately in the four-axis evaluation.